

Credit Card Default Prediction

Article reproduction and improved classification methodology

FINAL
PROJECT

Team Members

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Based on Ly, Schuller, Simanic, and Singh
UCI Default of Credit Card Clients dataset

1. Summary

This project predicts whether a credit card client will default on the next monthly payment. It reproduces the classification methodology from Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms by Thomson Ly, Rene Schuller, Katarina Simanic, and Sukhpreet Singh. The selected paper compares Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) using the public UCI Default of Credit Card Clients dataset.

The group first reproduced the article's cleaning, 80/20 split, standardization, random oversampling, three-fold GridSearchCV, and evaluation process. The reproduction confirmed the paper's main result: SVM was the strongest of the three models. The group then created an upgraded pipeline using one-hot encoding, fold-safe oversampling, and F1-focused model selection. Logistic Regression was also added as a fourth classification model.

Main result: the improved SVM reached a default-class F1-score of 0.534, compared with 0.514 in our reproduction and 0.520 in the article.

2. Motivation

Credit default prediction is an important classification problem because financial institutions need to identify clients with a higher risk of missing future payments. A useful model can support credit-risk review, but an incorrect prediction can create financial loss or unfairly classify a client. For this reason, model quality must be evaluated with more than accuracy.

We selected this paper because it addresses a relevant real-world problem with three supervised learning algorithms that are familiar enough to explain clearly in class. It also uses a public dataset, reports comparable classification metrics, and leaves useful opportunities for the group to improve preprocessing, validation, and minority-class model selection.

The project answers the following research questions:

- Can the main methodology and result of the selected article be reproduced?
- Which of Decision Tree, KNN, and SVM performs best for the default class?
- Can safer preprocessing and F1-focused tuning improve minority-class performance?
- Does adding Logistic Regression improve the comparison or provide a useful trade-off?

3. Research Paper Details

The primary paper is Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms. The authors use Python, scikit-learn, imbalanced-learn, StandardScaler, RandomOverSampler, and GridSearchCV. They optimize Decision Tree, KNN, and SVM and report accuracy, precision, recall, F1-score, and confusion matrices.

Item	Description
Authors	Thomson Ly, Rene Schuller, Katarina Simanic, and Sukhpreet Singh
Institution	University of Windsor
Objective	Compare supervised models for predicting next-month credit default.
Best reported model	SVM, with accuracy 0.76 and default-class F1-score 0.52.

Model	Hyperparameters tested	Article result
Decision Tree	criterion, splitter, max features, max depth	Accuracy 0.74; F1 0.41
KNN	neighbors, weights, distance metric	Accuracy 0.73; F1 0.39
SVM	kernel and C	Accuracy 0.76; F1 0.52

4. Dataset Details

The dataset is available from the UCI Machine Learning Repository (dataset ID 350). It contains credit limit, demographic information, payment status, bill amounts, payment amounts, and a binary target showing whether the client defaulted in the next month. The records were collected from credit card clients in Taiwan.

Item	Value
Original observations	30,000
Removed observations	399
Final observations	29,601
Input features	23
Default clients	6,605 (22.3%)
Target	Default next month: 0 or 1

The dataset has no missing values, but the article removes undocumented values in the EDUCATION and MARRIAGE variables. The target is imbalanced: approximately 77.7% of clients do not default and 22.3% default.

Feature groups and data types

Feature group	Columns	Source type / analytical role
Credit limit	LIMIT_BAL	Integer / numerical
Demographics	SEX, EDUCATION, MARRIAGE	Integer codes / categorical
Age	AGE	Integer / numerical
Payment status	PAY_0, PAY_2-PAY_6	Integer / ordered status
Bill amounts	BILL_AMT1-BILL_AMT6	Integer / numerical
Payment amounts	PAY_AMT1-PAY_AMT6	Integer / numerical
Target	default.payment.next.month	Integer / binary class

The original modeling shape is 30,000 rows by 23 input features, plus one target column. After applying the paper's cleaning rule, the modeling shape becomes 29,601 rows by 23 input features. Although every source column is stored as an integer, the improved approach treats demographic codes as categories because their numbers do not represent continuous quantities.

5. Data Preprocessing and Feature Engineering

Preprocessing was designed in two versions: an article-style workflow for reproduction and an improved workflow for the group's contribution. The test set remains untouched until final evaluation in both versions.

Stage	Article-style reproduction	Improved workflow and critical observation
Cleaning	Remove ID and 399 invalid category rows.	Use the same rule to keep a fair comparison.
Split	Stratified 80/20 train-test split.	Split before learning any transformation.
Categories	Integer codes remain in the matrix.	One-hot encode SEX, EDUCATION, and MARRIAGE.
Scaling	StandardScaler on the training data.	Scale numerical columns inside the pipeline.
Imbalance	Random oversampling before model tuning.	Oversample only inside each CV training fold.
Selection	Optimize cross-validation accuracy.	Optimize default-class F1-score.

The most important observations were the class imbalance, the undocumented category codes, and the risk of leakage when oversampling is performed before cross-validation. No synthetic feature was required; the main feature-engineering change was representing demographic codes with one-hot encoded indicator columns.

6. Steps Reproduced from the Paper

The reproduction follows the main steps described in the paper. A fixed random state of 42 is used in our code because the paper does not report its random seed.

Step	Implementation
1. Cleaning	Remove ID and 399 undocumented EDUCATION or MARRIAGE rows.
2. Split	Use a stratified 80% training and 20% testing split.
3. Scaling	Fit StandardScaler on the training data and transform the test data.
4. Balancing	Apply RandomOverSampler to the training set.
5. Tuning	Use three-fold GridSearchCV with accuracy as the article-style selection metric.
6. Testing	Evaluate the selected model once on the untouched test set.

6.1 Article hyperparameter workload

Model	Grid combinations	CV folds	Total fits
Decision Tree	1,932	3	5,796
KNN	114	3	342
SVM	6	3	18
Total	2,052	-	6,156

To make the notebook practical for classroom demonstration, the default execution uses the best article-style parameters already identified by the complete grid. Setting `RUN_FULL_ARTICLE_GRID` to `True` repeats all 6,156 fits. The full grids remain visible in the notebook for reproducibility.

6.2 Evaluation metrics

- Accuracy: proportion of all predictions that are correct.
- Precision: proportion of predicted defaults that are true defaults.
- Recall: proportion of true defaults detected by the model.
- F1-score: harmonic balance between precision and recall for the default class.

7. Contributions

The assignment gives the highest grading weight to contributions made after reproducing the paper. Our project provides both methodological improvements and an additional classification model.

Contribution	Why it matters
One-hot encoding	Treats SEX, EDUCATION, and MARRIAGE as categories instead of continuous values.
Fold-safe oversampling	Sampling occurs only inside each cross-validation training fold, reducing leakage risk.
F1-focused selection	Selects parameters using the minority-class objective instead of general accuracy.
Expanded parameters	Tests class weights, minimum leaf sizes, more K values, and SVM gamma settings.
Logistic Regression	Adds a fourth model not tested by the selected article.

7.1 Improved pipeline

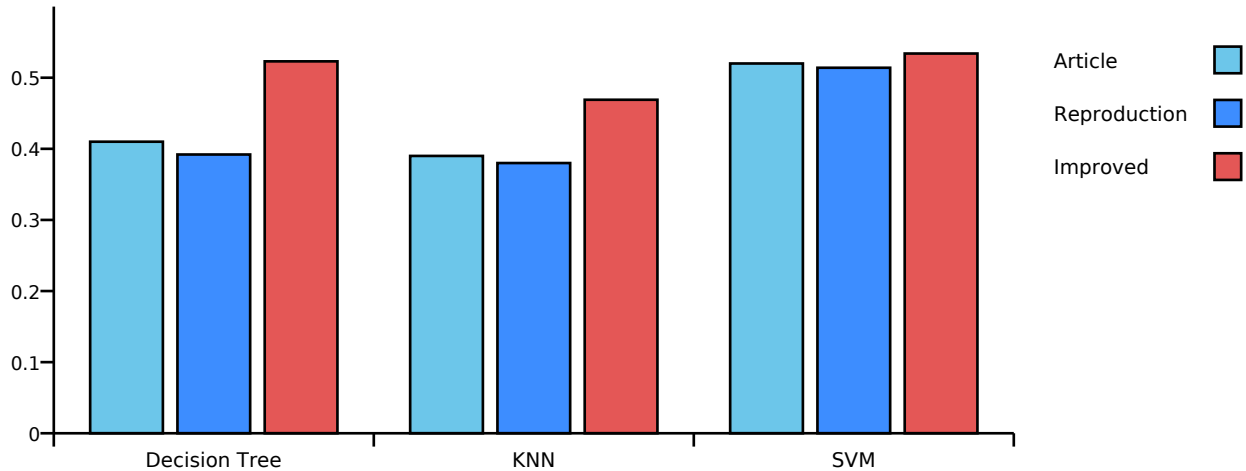
The improved pipeline contains a ColumnTransformer, StandardScaler for numerical variables, OneHotEncoder for categorical variables, RandomOverSampler, and the classifier. GridSearchCV receives the complete pipeline, so each validation fold receives preprocessing learned only from its own training fold. This is safer than oversampling the full training set before cross-validation.

7.2 Added Logistic Regression model

Logistic Regression was tested with C values of 0.01, 0.1, 1, and 10, with and without balanced class weights, using the liblinear solver. It obtained accuracy 0.692, precision 0.387, recall 0.647, and F1-score 0.484. Its recall was the highest of all improved models, but its overall F1-score remained below SVM.

8. Significant Improvements

Default-class F1-score comparison



Model	Article F1	Reproduced F1	Improved F1	Change vs reproduction
Decision Tree	0.410	0.392	0.523	+0.131
KNN	0.390	0.380	0.469	+0.089
SVM	0.520	0.514	0.534	+0.020
Logistic Regression	N/A	N/A	0.484	Additional model

Improved model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.768	0.483	0.572	0.523
KNN	0.681	0.373	0.632	0.469
SVM	0.776	0.498	0.575	0.534
Logistic Regression	0.692	0.387	0.647	0.484

Result highlight: SVM provided the strongest overall balance, while Logistic Regression detected the largest proportion of true default clients.

8.1 Interpretation of the Improvements

Our reproduction follows the same general pattern as the article. SVM is the strongest model and the reproduced F1-score of 0.514 is close to the paper's value of 0.520. Small differences are expected because the paper does not provide its random seed, exact data split, package versions, or every implementation detail.

The improved method increased the F1-score of all three original models. Decision Tree showed the largest increase, from 0.392 to 0.523. SVM improved more modestly, from 0.514 to 0.534, but remained the strongest overall model. This supports the argument that safer preprocessing and a more relevant selection metric can improve minority-class performance.

Logistic Regression did not replace SVM as the best model. However, its recall of 0.647 is important. It detects more default clients, but also creates more false positives, which reduces precision and accuracy. The preferred model therefore depends on the financial cost of each error.

Objective	Most useful result	Reason
Best overall balance	SVM	Highest F1-score and accuracy.
Detect more defaults	Logistic Regression	Highest recall.
Largest improvement	Decision Tree	F1 increased by 0.131.
Simple local patterns	KNN	High recall, but lower precision and accuracy.

9. Challenges

- Class imbalance: only 22.3% of clients are in the default class.
- Exact reproduction: the paper does not provide its random seed or all software details.
- Computational cost: the article grids require 6,156 model fits.
- Data leakage risk: oversampling before cross-validation can repeat observations across folds.
- Metric trade-offs: improving recall can reduce precision or accuracy.

9.1 Main learnings

- Accuracy alone is not enough for an imbalanced classification problem.
- Preprocessing and sampling should be fitted only on training data.
- The model-selection metric should match the real project objective.
- Scientific reproduction can be close without producing identical numbers.
- A model with higher recall may create more false-positive decisions.
- Clear documentation is necessary for reproducibility and explanation.

9.2 Individual responsibilities

Participant	Main responsibility
Simon	Motivation, research questions, and article overview.
Gabriel	Dataset, cleaning, preprocessing, and reproduction workflow.
Eduardo	Improved pipeline, Logistic Regression, tuning, and results.
Maria	Challenges, learning, conclusion, documentation, and Q&A.;

10. Conclusion and Future Scope

10.1 Limitations

- The dataset represents one Taiwanese bank and an older historical period.
- Random oversampling repeats minority observations and may increase overfitting.
- The project does not include probability calibration or a financial cost matrix.
- The final F1-score is moderate and is not sufficient for automatic credit decisions.
- The methodology was not tested on a second credit-risk dataset.

10.2 Future Scope

- Compare StandardScaler with MinMaxScaler.
- Test five-fold cross-validation and broader parameter ranges.
- Evaluate SMOTE and other imbalance-handling methods.
- Tune the classification threshold and calibrate probabilities.
- Compare Random Forest, XGBoost, and neural networks.
- Create a cost matrix for false positives and false negatives.
- Test the methodology on a newer or similar credit-risk dataset.

10.3 Conclusion

The project successfully reproduced the main methodology and result of the selected research paper. SVM remained the strongest model, confirming the article's conclusion. The improved pipeline raised default-class F1-score for Decision Tree, KNN, and SVM while reducing leakage risk and treating categorical variables more appropriately.

Adding Logistic Regression strengthened the contribution by introducing a fourth model. It achieved the highest recall but did not outperform SVM on F1-score. Overall, the work shows that a fair and well-documented evaluation process is as important as selecting a winning model.

References

1. Ly, T., Schuller, R., Simanic, K., and Singh, S. Predicting Default of Credit Card Clients Using Three Supervised Machine Learning Algorithms. University of Windsor.
2. UCI Machine Learning Repository. Default of Credit Card Clients. Dataset ID 350.
<https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>
3. Yeh, I.-C., and Lien, C.-H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. Expert Systems with Applications, 36(2), 2473-2480.
<https://doi.org/10.1016/j.eswa.2007.12.020>
4. DS 301 Final Project Details. Course assignment and grading rubric.

Repository Deliverables

Item	Location
README	README.md
Dataset	data/default_of_credit_card_clients.xls
Machine learning notebook	models/credit_default_ml_simplificado.ipynb
Project report	Project_Report.pdf
Presentation	Credit_Card_Default_ML_Presentation.pptx
Research paper	references/MLEArticle.pdf